

U2S2: Agglomerative Algorithms

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GAS: Generalized Agglomerative Scheme

Initialization:

- Choose $R_1 = \{C_i = \{x_i\}, i=1, \dots, N\}$
- $t=1$

This is R_1 where each data point is its own cluster

Repeat:

- $t=t+1$
- Among all pairs on $(C_i, C_j) \in R_{t-1}$ find (i, j) that satisfies:
 - If g is the dissimilarity function $g(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$
 - (or max if we are working on similarity)
- Define $C_q = C_i \cup C_j$ and produce a new clustering
- $R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$
- (we get rid of the ones we merged and add the merged set)

Main idea: on each iteration the new R_t will only differ on 1 cluster as it will only merge 2 we found closest. This allows for the one merged in all successive clusters

$$R_t = \{R_{t-1} - \{C_i, C_j\} \cup \{C_q\}\}$$

AKA once 2 things are in the same cluster they will be for all the next microclusters.

PATTERN MATRIX:

L-dimension

This means $D(X)$ is the $N \times N$ matrix whose i th row is the transposed i th vector of X

Similarity / Dissimilarity matrix $P(X)$ is an $N \times N$ matrix whose (i, j) element is the distance between the i th and j th elements of X we may call it PROXIMITY MATRIX to include both cases.

In general the PROXIMITY MATRIX is symmetric

- Moreover:
- If P is a sim diagonal = 1
 - If P is a diss diagonal = 0

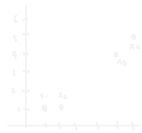
Weakness of the Method:

- If a poor clustering choice is made earlier it is not possible to recover
- The algorithm goes over the data N^2 times but the actual big O depends on how we are defining the metric & method the distance between 2 clusters.

EXAMPLE OF PATTERN & PROXIMITY

let $X = \{x_i, i=1, \dots, 4\}$ with

- $x_1 = (1, 1)$
- $x_2 = (2, 1)$
- $x_3 = (5, 4)$
- $x_4 = (6, 5)$



PATTERN

$$D(X) = \begin{bmatrix} 1,1 \\ 2,1 \\ 5,4 \\ 6,5 \end{bmatrix}$$

Dissimilarity matrix

L_1 Manhattan	L_∞ Chebyshev
$x_1 \quad x_2 \quad x_3 \quad x_4$	$x_1 \quad x_2 \quad x_3 \quad x_4$
$x_1 \quad 0 \quad 1 \quad 7 \quad 4$	$x_1 \quad 0 \quad 1 \quad 4 \quad 5$
$x_2 \quad 1 \quad 0 \quad 6 \quad 8$	$x_2 \quad 1 \quad 0 \quad 5 \quad 4$
$x_3 \quad 7 \quad 6 \quad 0 \quad 1$	$x_3 \quad 4 \quad 5 \quad 0 \quad 1$
$x_4 \quad 4 \quad 8 \quad 1 \quad 0$	$x_4 \quad 5 \quad 4 \quad 1 \quad 0$

THE DENDROGRAM

A threshold dendrogram is a way to represent

effect to some cluster formed

• Cut a dendrogram and you get a partition.